

DBMS — ACID Properties

Core Transaction Concepts for GATE, UGC-NET & University Exams

REVISION GUIDE

ACID represents the four fundamental properties that a transaction management system in a Database Management System (DBMS) must satisfy to guarantee reliability and data integrity.

Letter	Property	Core Meaning	Memory Trick
A	Atomicity	All or None execution	All or None
C	Consistency	Valid state → Valid state	Correct State
I	Isolation	Parallel acts like serial execution	Independent Execution
D	Durability	Committed changes are permanent	Durable / Permanent

1. Atomicity — All or None

Definition: Atomicity treats a transaction as a single indivisible unit. Either all operations succeed, or none take effect. If a failure occurs (even at the 99th step), the system performs a **ROLLBACK** to undo all partial changes.

Fund Transfer Example (A → B)

```
READ(A)
A = A - 1000
WRITE(A)
[SYSTEM FAILURE HERE]
READ(B)
B = B + 1000
WRITE(B)
COMMIT
```

Prevents state where ₹1000 is deducted from A but not credited to B.

Rollback Process

Transaction Failure / Crash



ROLLBACK Triggered



Undo Changes (Original State Restored)

⚠ Exam Trap: Atomicity does *not* mean simultaneous execution. It means the *final effect* is treated as all-or-nothing.

★ Golden Rule: Atomicity = ALL or NONE

2. Consistency — Valid State → Valid State

Definition: Ensures a transaction brings the database from one consistent state to another consistent state, preserving all integrity constraints (e.g., total sum of balances remains constant).

Account	Before Transfer	After Transfer
Account A	₹2,000	₹1,000
Account B	₹3,000	₹4,000
Total Balance	₹5,000	₹5,000

ATM Withdrawal Scenario

Account Balance = ₹10,000. Request = ₹5,000 withdrawal.

- **Consistent:** ₹5,000 deducted AND ₹5,000 cash dispensed.
- **Inconsistent:** ₹5,000 deducted BUT no cash dispensed (system must trigger rollback).

★ Golden Rule: Consistency = Database remains valid before and after execution.

3. Isolation – Concurrent Execution Without Interference

Definition: When multiple transactions run concurrently, isolation ensures their operations do not improperly interfere with each other. The net outcome should behave as if transactions executed serially (one after another).

Concurrent / Interleaved Schedule

```
T1: Read(A)
T2: Read(A)
T1: Write(A)
T2: Write(A)  (Interleaved)
```

Executed in parallel for efficiency.

Conceptual Serial Equivalent

```
T1 Complete → T2 Start
OR
T2 Complete → T1 Start
```

Achieved via Conflict / View Serializability.

★ **Golden Rule: Isolation = Concurrent transactions execute independently.**

4. Durability – Permanent Changes

Definition: Once a transaction executes a **COMMIT**, all changes are permanently saved in non-volatile storage (e.g., hard disk). The changes will survive subsequent system crashes or power failures.

Transaction Execution → **COMMIT** → Written to Disk → **System Crash/Restart** → **Data Preserved**

★ **Golden Rule: Durability = COMMITTED changes stay permanent.**

All Four Properties in One Transfer Scenario

Scenario: Transfer ₹1000 from Account A (Initial: ₹2000) to Account B (Initial: ₹3000).

A – Atomicity

Either full ₹1000 transfers or complete rollback occurs if system fails midway.

C – Consistency

Total sum remains ₹5000 both before (2000+3000) and after (1000+4000) transfer.

I – Isolation

Other active transactions cannot view or overwrite dirty intermediate balances.

D – Durability

After COMMIT, balances A=1000 & B=4000 are saved permanently on disk.

Critical Exam Distinctions & Quick Summary

Atomicity	Durability	Consistency	Isolation
All or None execution.	Permanent persistence.	State validity & rules.	Concurrent interference.
Applies during execution.	Applies after COMMIT.	Focuses on data correctness.	Focuses on schedule execution.
Uses ROLLBACK on failure.	Saves log/data to non-volatile disk.	Example: Total money balance.	Example: Interleaved T1/T2.

10-Second Revision Summary

A → Atomicity (All or None) | **C** → Consistency (Valid State → Valid State) | **I** → Isolation (No Interference) | **D** → Durability (Committed Changes Persist)